Task Analyzer
Phase 2 technical report

DRAFT - NOT READY FOR ACADEMIC SUBMISSION

Prepared on 2026-10-16 (America/Sao_Paulo). Repository owner: BudaBecker. The implementation and local checks are complete. Actual screenshots, human review and the required five-minute video remain pending.

Published implementation

Public repository: BudaBecker/sdd-task-analyzer

Phase 2 pull request #4 - open for review

Phase 2 source tree

Approved Phase 1 sources identified by the user (.specs/)

Delivered artifacts

Artifact	Purpose
src/task_analyzer.py	Typed Task records, analyze_tasks and TaskValidationError.
tests/test_harness.py	30 success, failure and boundary cases.
specs/task_analyzer_spec.md	Batch signature, reconciliation and acceptance examples.
CONTEXT_RULES.md	AI governance, evidence limits and approved boundaries.
pyproject.toml / uv.lock / requirements.txt	uv remains authoritative; requirements is an exported course artifact.

Contract reconciliation

The existing .specs/ features cover persistence and lifecycle. The Phase 2 contract explicitly adds the batch boundary; it does not claim those files already contained the challenge's numbered examples. Counts, elapsed duration and overdue eligibility preserve approved product examples. Priority grouping is an explicit academic input, not a new persistent task field. Numeric empty averages do not change the desktop's N/A presentation rule.

Verification and behavior

Actual local command results

Text record of executed checks, not a screenshot. The complete record is versioned in docs/phase-two/verification.txt. Tests use disposable paths; no real database was initialized or modified.

```
uv run python -m pytest tests/test_harness.py -v
30 passed in 0.07s

uv run python -m pytest -q
518 passed, 2 warnings in 61.95s (0:01:01)

uv run ruff format --check src tests
39 files already formatted

uv run ruff check src tests
All checks passed!

uv run python -m mypy --strict
Success: no issues found in 12 source files

uv pip check
All installed packages are compatible

uv build
Source distribution and wheel built successfully
```

The full suite consists of 487 existing server cases, 30 batch cases and one new installed-wheel batch smoke test outside the checkout. Two third-party deprecation warnings remain in Starlette/httpx and anyio integration.

Acceptance evidence

Scenario	Expected and verified outcome
1 - successful analysis	5 total, 3 completed, 2 pending; mean 3.0 hours; priority 4.0 and non-priority 2.5 hours; overdue rate 50.0%.
2 - invalid chronology	Completion before original creation raises TaskValidationError without a partial result or input mutation.
Empty / pending-only	Every average and rate is 0.0; counts retain their meaning.
Time and eligibility	UTC elapsed durations across DST; cutoff equality is late; pending/deleted tasks are excluded from the appropriate populations; Apia skipped-date rule retained.

These checks do not establish Ubuntu compatibility, Tailscale access, a metrics API or a completed Windows desktop application.

AI governance and completion

Prompt history and critical validation

The actual session instructions were to compare the intermediate challenge with the repository, execute the Phase 2 work with commits and pushes, and use .specs/ as the Phase 1 source. This is a summary, not a fabricated verbatim prompt transcript.

Context rules kept the implementation typed, read-only and governed by the approved time rules. The code reuses the existing deadline resolver. Tests check the 2-hour/4-hour mean, unrounded overdue rates, chronology failures and empty denominators. The first lint pass found an overlong docstring; it was corrected. Packaging was tested from an installed wheel to catch missing-module failures that source-tree tests could hide.

Human review is pending. Passing tests support the cases executed; they are not proof that all AI output is correct or that hallucinations are impossible.

Required final evidence

Item	Current status / next action
Terminal screenshot	Pending. Capture the actual harness command and 100%-passing summary.
GitHub tree screenshot	Pending. Capture the published Phase 2 branch, including required artifacts.
Pull request screenshot	Pending. Capture PR #4 with its changes and verification. Human review remains pending.
Five-minute YouTube video	Pending. Record horizontally, with visible camera and audio; use Unlisted or Public visibility, not a Short. No URL has been supplied.
Phase 1 Docs link / student details	Confirm whether the rubric accepts the linked .specs/ source or requires a separate Google Docs URL. Add any required academic identification.
salaonline submission	Not performed. Complete the evidence, update this PDF and review before submission.

Recording outline and source material

Submission guide, five-minute outline and remaining checklist

No deployment, real-database operation or PR merge is part of this delivery. The repository is public and renamed to match the required sdd-task-analyzer name; the local checkout directory retains its original name.